EE4-29
EE9-CS3-2
EE9-FPN1-04

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

OPTIMIZATION

Thursday, 29 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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OPTIMISATION

1. The Lagrangian Sufficiency Theorem for the optimization problem

$$P: \begin{cases} \min\limits_{x} \ f(x), \\ g(x) = 0, \end{cases}$$

states that if there exist $x^\star$ and $\lambda^\star$ such that $x^\star$ is feasible and

$$L(x^\star, \lambda^\star) \leq L(x, \lambda^\star),$$

where $L(x,\lambda) = f(x) + \lambda^\top g(x)$ is the Lagrangian of the problem, then $x^\star$ is an optimal solution of problem P.

Consider problem P with $x = (x_1, x_2, x_3)$,

$$f(x) = x_1 - x_2 + 2x_3, \qquad g(x) = \begin{bmatrix} x_1 + x_2 + x_3 \\ x_1^2 + x_2^2 - 1 \end{bmatrix}.$$

One wishes to use the Lagrangian Sufficiency Theorem to determine optimal solutions for this problem.

a) Write first order necessary conditions of optimality and determine candidate optimal solutions. [6 marks]

b) Determine conditions on the Lagrange multiplier such that the Lagrangian is bounded from below. Hence, minimise the Lagrangian with respect to x_1, x_2 and x_3 for fixed λ and explicitly compute the minimiser $x^\star(\lambda)$ of the Lagrangian as a function of λ.
(Hint: minimise independently with respect to x_1, x_2 and x_3.) [8 marks]

c) Show that there exists a unique $\lambda = \lambda^\star$, satisfying the conditions in part b), such that $x^\star(\lambda^\star)$ is feasible. Hence argue, using the Lagrangian Sufficiency Theorem, that $x^\star(\lambda^\star)$ (where the function $x^\star(\lambda)$ has been determined in part b)) solves the considered problem. Show that the obtained optimal solution coincides with one of the candidate optimal solutions computed in part a).
[6 marks]

2. Consider sending data of rate r over a choice of two parallel queues, denoted as queues 1 and 2, each with a processing rate of C. Assume $C > 1$ and $r + 1 < 2C$.

The data is split into separate streams of rates x_1 and x_2, where data of rate x_i is sent into queue i. The first queue has no additional traffic, while the second queue serves an additional traffic stream of rate 1.

Suppose the average number of packets in a queue is equal to $\dfrac{f}{C-f}$, where f is the total arrival rate to the queue.

We want to determine x_1 and x_2 to minimize the total average number of packets in the system. This can be formulated as the optimization problem

$$P: \begin{cases} \min\limits_{x_1,x_2} \dfrac{x_1}{C-x_1} + \dfrac{x_2}{C-(x_2+1)}, \\ r \le x_1 + x_2, \\ x_1 \in [0,\, C),\ x_2 \in [0, C-1). \end{cases}$$

Instead of solving this problem, one can consider the simplified problem

$$P_s: \begin{cases} \min\limits_{x_1,x_2} \dfrac{x_1}{C-x_1} + \dfrac{x_2}{C-(x_2+1)} \\ r = x_1 + x_2, \end{cases}$$

and then impose the constraints

$$x_1 \in [0,\, C), \qquad x_2 \in [0, C-1) \qquad (\star)$$

a posteriori.

Assume that $r = 2$ and $C = 2$.

a) Use the constraints elimination method to determine the solution for the problem P_s. Check that the solution satisfies the constraints $(\star)$. [4 marks]

b) Write the necessary conditions of optimality for the problem P_s. [2 marks]

c) Using the conditions in part b) determine candidate optimal solutions for the problem P_s.
(Hint: there are 2 candidate solutions.) [8 marks]

d) Show that of the 2 candidate solutions determined in part c) only one satisfies the constraints $(\star)$. Show that this coincides with the solution determined in part a). [2 marks]

e) Using second order sufficient conditions of optimality show that the candidate optimal solution determined in part d) is a local minimiser for problem P_s. [4 marks]

3. Consider the optimization problem

$$\begin{aligned}
&\min_{x_1,x_2,x_3} \; x_1^2 + x_2^2 - x_3^2, \\
&x_1 - x_2 - x_3 = 0, \\
&-x_1 + x_2 - \frac{1}{10} \leq 0, \\
&-x_2 + (x_3 + x_2 - 1)^4 \leq 0.
\end{aligned}$$

a) Show that by solving the equality constraint in the variable x_3 one obtains the reduced problem

$$\begin{aligned}
&\min_{x_1,x_2} \; 2x_1x_2, \\
&-x_1 + x_2 - \frac{1}{10} \leq 0, \\
&-x_2 + (x_1 - 1)^4 \leq 0.
\end{aligned}$$

[2 marks]

b) By plotting the admissible set and the level lines of the reduced problem determine its solution. Show, in addition, that all points are regular points for the constraints. [6 marks]

c) Write the Lagrangian for the reduced problem and show that the solution determined in part b) satisfies the necessary conditions of optimality for some selection of the multipliers. Explicitly compute the optimal multipliers. Show that the condition of strict complementary holds at the optimal point. [6 marks]

d) Show that second order sufficient conditions of optimality do not allow determining that the optimal solution obtained in part b) is a local minimizer. [4 marks]

e) Discuss alternative ways to assess, analytically, local optimality of the point determined in part b).
(Hint: please provide a short answer.) [2 marks]

4. Consider the so-called root-finding problem, that is the problem of solving the equation

$$f(x) = 0,$$

with $x \in \mathbb{R}$. Assume that f is twice differentiable and denote with f' and f'' its first and second derivative, respectively.

Newton's method for the solution of this problem is given by the well-known iteration

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \qquad (N)$$

Newton's method can be improved in various ways. For example, Steffensen's method, described by the iteration

$$x_{n+1} = x_n - \frac{f(x_n)}{g(x_n)}, \qquad g(x_n) = \frac{f(x_n + f(x_n)) - f(x_n)}{f(x_n)}, \qquad (S)$$

does not require the evaluation of the derivative of f, whereas the family of Chebyshev–Halley methods, described by the iteration

$$x_{n+1} = x_n - \left[1 + \frac{1}{2}\ \frac{L_f(x_n)}{1 - \lambda L_f(x_n)}\right] \frac{f(x_n)}{f'(x_n)}, \qquad L_f(x_n) = \frac{f''(x_n) f(x_n)}{(f'(x_n))^2}, \qquad (CH)$$

with $\lambda \geq 0$, yields cubic convergence speed.

Consider the problem of determining the square roots of 2. This problem can be posed as the problem of finding the positive root of the function $f(x) = x^2 - 2$. Let $x_0 = 1$ be the initial approximation of the solution of this equation.

a) Write Newton's iteration for the considered problem in the form

$$x_{n+1} = N(x_n).$$

Determine explicitly the function N. Hence, run three iterations of Newton's method with initial point x_0. [2 marks]

b) Write Steffensen's iteration for the considered problem in the form

$$x_{n+1} = S(x_n).$$

Determine explicitly the function S. Hence, run three iterations of Steffensen's method with initial point x_0. [4 marks]

c) Write the Chebyshev–Halley iteration, with $\lambda = 1/2$, for the considered problem in the form

$$x_{n+1} = CH(x_n).$$

Determine explicitly the function CH. Hence, run three iterations of the Chebyshev–Halley method (with $\lambda = 1/2$) with initial point x_0. [4 marks]

d) Compare the convergence properties of the sequences determined in parts a), b), and c). Explain why it is reasonable to expect that the Chebyshev–Halley iteration has the fastest convergence speed, while Steffensen's iteration has the slowest convergence speed. [2 marks]

e) By plotting, on the xy-plane, the functions $y = N(x)$, $y = S(x)$, $y = CH(x)$, determined in parts a), b) and c), and the function $y = x$, all for $x \geq 0$, show that the following holds.

i) The functions $N(x)$, $S(x)$, $CH(x)$, and x are such that

$$\sqrt{2} = N(\sqrt{2}) = S(\sqrt{2}) = CH(\sqrt{2}).$$

Explain why this is the case. [2 marks]

ii) For every $x_0 > 0$ Newton's iteration yields a sequence $\{x_n\}$ which converges to $\sqrt{2}$. For every $x_0 > \sqrt{3} - 1$ Steffensen's iteration yields a sequence $\{x_n\}$ which converges to $\sqrt{2}$. For every $x_0 > 0$ the Chebyshev–Halley iteration yields a sequence $\{x_n\}$ which converges to $\sqrt{2}$. [3 marks]

iii) Discuss what happens to the sequence generated by Newton's iteration for $x_0 = 0$; to the sequence generated by Steffensen's iteration for $x_0 = \sqrt{3} - 1$ and for $x_0 \in [0, \sqrt{3} - 1)$; and to the sequence generated by the Chebyshev–Halley iteration for $x_0 = 0$. [3 marks]